

MUHAMMAD TAHA CHEEMA

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EDUCATION

University of Waterloo

Waterloo, Canada

Master of Mathematics in Computer Science (Thesis) **Grade: 99.5/100**

Jan 2026 – Present

- Awards: International Master's Award of Excellence.
- Coursework: Adv. Topics in SE - Large Language Models for SE, Adv. Topics in SE - SE for Big Data and AI.

Lahore University of Management Sciences (LUMS)

Lahore, Pakistan

Bachelor of Science in Computer Science **GPA: 3.97/4.0**

Sep 2021 – May 2025

- Awards: Graduated with High Distinction, Dean's Honor List (2021-2025), LUMS Merit Scholarship (2021-2025)

Stanford University

California, USA

Exchange Quarter **GPA: 4.0/4.3**

Jun 2023 – Aug 2023

- Coursework: CS 229 - Machine Learning, ENGR 145 - Technology Entrepreneurship.

TECHNICAL SKILLS

Languages: Python, C/C++, Golang, TypeScript, JavaScript, SQL, HTML, CSS, Haskell

AI/ML & Data: LLM pipelines & orchestration, vector & semantic search, Meilisearch, Milvus, Hugging Face, OpenCV, RAG

Tools & Web: React, Redux, Next.js, Recoil, Material UI, Motion, tRPC, Prisma ORM, Git, Docker, AWS, Railway, Supabase

EXPERIENCE

University of Waterloo

Waterloo, Canada

Graduate Research Assistant

Jan 2026 – Present

- Architected an **over-the-air testbed** for non-terrestrial (NTN) 5G. **Led key technical decisions** (software stack's integration with radio hardware, channel emulator choice) and **scoped 6 NTN optimization avenues** through a literature review.
- Mapped the design onto the open-source **OpenAirInterface** 5G stack and enabled **real-time KPI measurement** by porting the hardware team's custom channel-impairment models to its simulator package in C.

GreatFrontEnd

Singapore

Software Engineer

Aug 2025 – Jan 2026

Software Engineering Intern (Part-time)

Feb 2025 – Apr 2025

- Led AI engineering** on their new product, Socialmon, an AI companion for brands to create viral marketing ideas.
- Developed a pipeline to generate editable ad and social-media post layouts from natural-language prompts. **Extracted high-fidelity JSON representations** of public viral posts via background masking and **Gemini's image segmentation**, then used them as **few-shot examples** to **improve the aesthetic quality of the LLM-generated HTML designs**.
- Programmed and **deployed a scalable, real-time artifact search engine** on Railway utilizing **vector and lexical search** and integrated a custom **re-rank algorithm** to ensure semantic variation among top search hits. Scaled to **118k data points with search latencies under 50ms** on a test dataset, proving deployment-readiness.
- Built an in-browser **executable JS IDE** for their interview preparation platform by **running code in a sandboxed iframe**, enabling learners to practice their coding without navigating away from the web page. **Increased time-on-page by 1.4x**.

AI Centre for Educational Technologies

Singapore

Software Engineering Intern

May 2024 – Aug 2024

- Engineered SoftMark, a **productionized digital grading platform deployed at NUS** and secondary schools in Singapore. Created a 2D grid allocation component to **increase fairness in grading workload distribution among teachers**.
- Implemented a trie-based algorithm over open-source OCR** to detect question numbering on answer sheets, **doubling teachers' productivity** by filtering unattempted questions.

Distributed and AI Systems Lab, LUMS

Sep 2024 – Apr 2025

Undergraduate Research Assistant

Lahore, Pakistan

- TweakLLM: A Routing Architecture for Dynamic Tailoring of Cached Responses**. Link: [arXiv:2507.23674](https://arxiv.org/abs/2507.23674).
- Programmed and benchmarked an **adaptive caching architecture for LLM systems** that uses a lightweight model to dynamically adapt cached responses to similar user queries, **cutting compute costs by up to 68%** on real-world user-LLM interaction datasets.

PROJECTS

Digitizing Partition-Era Abduction Registers

Oct 2025 – Dec 2025

Research Contract - School of Humanities and Social Sciences

LUMS

- Orchestrated Google's **Gemini API to digitize 37,210 entries** from 1940s India-Pakistan partition abduction registers, turning the historical archive into an **indexable, searchable** dataset for learning and research.

Optimizing Vector Databases for Audio-Based RAG LLMs

Feb 2024 – May 2024

Course Research Project – CS678 Topics in Internet Research

LUMS

- Benchmarked Milvus, Weaviate, and Qdrant on **audio data embeddings** created from the Emotional Speech Dataset. Discovered an optimal configuration that achieved **4x better-than-random** speaker and emotion detection accuracy with **sub-5ms latency**, proving the potential for **real-time audio-based user interaction with LLMs**.

Improving Federated Learning for Causal Graph Discovery

Sep 2023 – May 2024

Directed Research

LUMS

- Identified a fundamental privacy-accuracy trade-off when replicating SOTA results for causal graph discovery, and **proposed an LLM-augmented, chain-of-thought discovery framework** to enhance performance. Mentored by Dr. Ihsan Ayyub Qazi.

ACHIEVEMENTS AND HONORS

Jun 2026: Third Place, University of Waterloo Future Cities Innovation Challenge

University of Waterloo

2025 – 2027: International Master's Award of Excellence valued at C\$12,500

University of Waterloo

2023: 70% Tuition Waiver Award

Stanford University

Feb 2020: Champions, Lahore Regional - National Science Bowl Pakistan